

Dhananjayan M

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EXECUTIVE SUMMARY

AI and Data Science student with hands-on experience in Machine Learning, Deep Learning, Data Analytics, and Retrieval-Augmented Generation systems. Proficient in Python, scikit-learn, TensorFlow, SQL, and data preprocessing. Passionate about building intelligent systems and applying AI techniques to solve real-world problems.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+)
Data & ETL	Pandas, NumPy, Matplotlib — preprocessing, ETL pipelines, aggregation
Databases	MySQL, SQLite — complex queries, joins, indexing, normalization, schema design
Frameworks	Django (REST APIs, MVT, ORM), React (hooks, components)
ML & AI	scikit-learn, TensorFlow, YOLOv8 — classification, regression, model evaluation
Core Concepts	Data Structures & Algorithms, OOP, ETL Design, Relational Modelling
Tools	VS Code, GitHub, Roboflow

EDUCATION

B.E. – Artificial Intelligence & Data Science

Coimbatore Institute of Engineering and Technology, Coimbatore | **CGPA: 8.48**

HSC (12th) – Tamil Nadu State Board | 78.6% | 2023

Sri Venkateshwara Matric Higher Secondary School, Tamil Nadu

PROJECTS

Tech Stack: Python, Django, LangChain, FAISS, MySQL, JavaScript, Chart.js, OpenCV

AI Document Question Answering System using RAG

- Built an AI-powered question answering system using LangChain and FAISS for semantic document retrieval.
- Implemented semantic search and context-aware question answering by retrieving relevant document content and generating responses based on uploaded knowledge sources.
- **GitHub:** [ChatWithPDF-RAG](#)

Sales Data ETL Dashboard

- Architected an end-to-end ETL pipeline to ingest CSV sales data, clean and transform it using Pandas, and load structured records into MySQL with duplicate handling.
- Developed an analytics dashboard with Chart.js to visualize KPIs — revenue trends, order counts, and category-wise performance using SQL aggregation queries.
- **GitHub:** [Sales data ETL dashboard](#)

Dynamic Form Builder

- Designed a relational MySQL schema to store dynamic form structures and responses; implemented data export to Excel and PDF by jQuery.

- Optimized multi-table relationships using foreign keys, joins, and indexed queries for fast retrieval and reporting.
- **GitHub:** [Dynamic Form Builder](#)

EB Bill Calculation

- Executed a rule-based data transformation engine in Python/Django to compute tiered electricity tariffs; stored multi-user monthly consumption history in MySQL.
- Architected SQL-driven reporting with GROUP BY aggregations to display per-user billing history and trends.
- **GitHub:** [EB-Bill calculation](#)

RailConnect – Train Ticket Booking System

- Modelled a normalized relational MySQL database for routes, stations, pricing classes, and bookings; enforced distance-based dynamic pricing logic at query time.
- **GitHub:** [Train Ticket Booking](#)

CERTIFICATIONS & AWARDS

Python Full Stack Web Development

TCDS Academy | 6 Months

- Completed intensive full-stack training covering Django (MVT, ORM, REST APIs), MySQL, HTML5/CSS3/Bootstrap, and Git across 5 hands-on projects.

Drone Phenomics Hackathon — Fairgrounds & Gates Foundation

- Architected an automated data selection pipeline using Databricks and Roboflow for drone crop imagery, gaining exposure to distributed data processing concepts.

Game Development — 1st Prize | ₹5,000 Cash Award

- Developed a Super Mario-style game in Godot and won 1st Prize at the inter-college competition.

CONFERENCE PRESENTATION

An Explainable AI-Based Vehicle Component Recognition and Bilingual Learning System for Automation in Process Industries

National Conference on Sustainable Smart Manufacturing and Automation (SSMA 2026)

Presenter | March 2026 | Certificate ID: SSMA 139

An AI-Based Smart Farming Decision Support System for Precision Agriculture

International Conference on Sustainable Energy Governance and Climate Cooperation (SEGCC) – ECOFEST'26, Tamil Nadu Agricultural University (TNAU), Coimbatore

Presenter | May 2026